Task 2 — Loan Approval Prediction

Alfido Tech Data Science Internship | Project report

Task 2: Loan Approval Prediction

Dataset and methodology

• Source: locally available CSV in `data/raw/`.

• Findings describe this dataset only; they are not automatically generalizable.

Findings

• Rows: 4,269; columns: 13.

• Duplicate rows: 0.

• Cells missing values: 0.

• Target class counts: approved=2,656; rejected=1,613.

• Random Forest held-out test accuracy: 0.9754 (97.5%); test records: 854.

• Classification report:

precision recall f1-score support

approved 0.98 0.98 0.98 531

rejected 0.97 0.97 0.97 323

accuracy 0.98 854

macro avg 0.97 0.97 0.97 854

weighted avg 0.98 0.98 0.98 854

• Confusion matrix saved to `outputs/task2_loan_confusion_matrix.csv` and PNG.

• This is a baseline on a public sample dataset. It does not establish real-world lending suitability, fairness, calibration, or causal validity.

Outputs

Charts and supporting CSV files are saved in outputs/.

Recommendations and next steps

• Compare logistic regression and a tree-based model using consistent train/test splits.

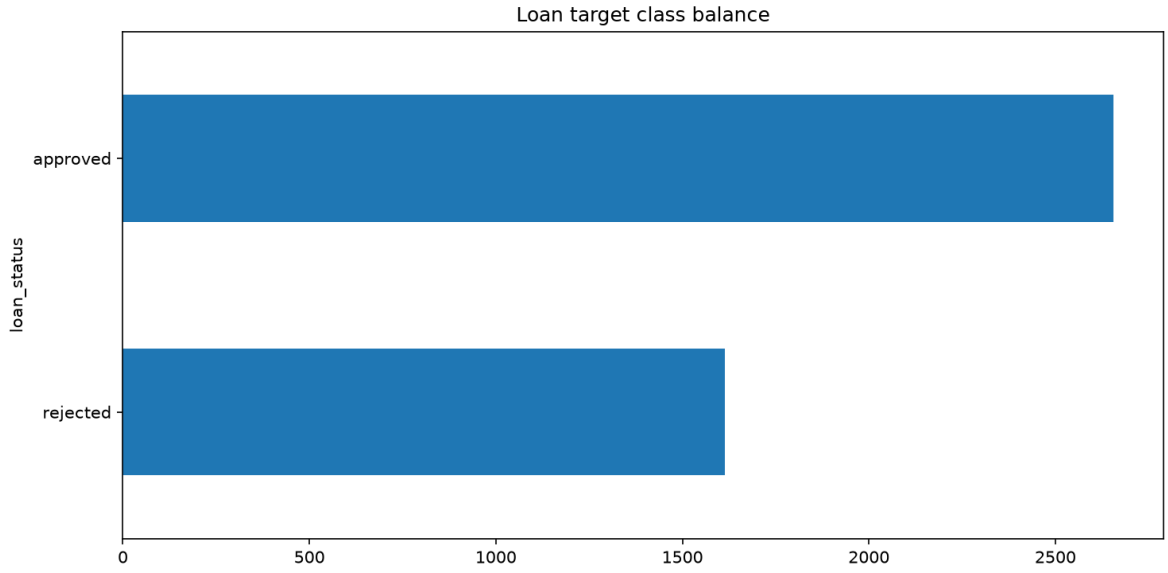
• Evaluate precision, recall, F1 and ROC-AUC, not accuracy alone.

• Inspect class imbalance and false approval/false rejection costs.

• Choose a decision threshold only after validation and cost analysis.

• Treat these results as educational experiments, not real-world lending advice.

Task2 Loan Class Balance



Task2 Loan Confusion Matrix

